

Social orderings for the assignment of indivisible objects [☆]

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Abstract

In the assignment problem of indivisible objects with money, we study social ordering functions which satisfy the requirement that social orderings should be independent of changes in preferences over infeasible bundles. We combine this axiom with efficiency, consistency and equity axioms. Our result is that the only social ordering function satisfying those axioms is the leximin function in money utility.

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1. Introduction

A set of objects have to be assigned to a group of agents. All objects are desirable but each agent may consume at most one object. Let us think of apartments in a housing complex, seats at a concert, parking lots, tasks in a board of directors, etc. Since preferences may differ among agents, monetary compensations are allowed for those who do not receive any objects or who receive an object they would be willing to exchange for another object. What could be an equitable way of assigning objects among those agents, and how should the compensations be computed?

The standard approach to this problem consists in looking for allocation rules (see Thomson [25] for a survey). An allocation rule specifies which feasible assignments are the most

[☆] I began this research while visiting the Institute for Advanced Study (Princeton). This paper owes a lot to the stimulating research atmosphere at the Institute.

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desirable as a function of the parameters of the problem, that is, the set of agents, the set of objects, and the agents' preferences over object–money bundles.

In many circumstances, however, it is not sufficient to know which assignments are the most desirable. The main reason comes from information and incentive constraints, which often prevents the planner from reaching first best alternatives. Given agents' incentive to misreport their preferences, given the information they own about each other, the set of reachable alternatives may have very different shapes, and typically does not contain the most desirable alternatives. Incentives are not the only reason why first best allocations may not be available. For instance, there may exist a status quo and reallocating objects may be too costly, so that only changes in monetary compensations are possible. Or there may be other feasibility constraints. As a result, an allocation rule should, ideally, identify the desirable assignments for each and every relevant set of feasible assignments.

An alternative approach consists in looking for social ordering functions. A social ordering function specifies a complete ranking of the feasible allocations as a function of the parameters of the problem. Having defined a social ordering allows the planner to solve the normative problem in all cases, by simply maximizing the ordering under the relevant incentive and feasibility constraints. This paper deals with social ordering functions.

Several recent studies of fairness in economic domains have succeeded in building social ordering functions based on fairness properties (see Fleurbaey [6], Fleurbaey and Maniquet [7, 8], and Maniquet and Sprumont [14] and [15]). That undertaking is clearly related to the social choice tradition of defining social welfare in such models. The social choice literature, however, has focused on properties of independence with respect to changes in preferences, and it has mainly uncovered impossibility results (see Le Breton and Weymark [13] for a survey). In the recent literature on social ordering functions, independence properties are weakened and the emphasis shifts towards fairness properties.

This paper is at the intersection of those two branches of literature. Indeed, we combine an independence property borrowed from social choice literature with consistency and fairness properties borrowed from fair social ordering literature and still obtain a possibility result (moreover, we characterize a unique social ordering function). The independence property we study requires that the social orderings be independent of preferences over infeasible alternatives (cf. Plott [19]; see Le Breton and Weymark [13] for a survey on applications of the Plott axiom in economic domains). This amounts to requiring that preferences over bundles containing objects that are not available should not matter, which seems extremely natural.

Our first, intermediary, result is a characterization of the social ordering functions that satisfy efficiency, our independence property and consistency. This result makes clear why traditional social choice theory and the fair allocation literature can be reconciled in the assignment of indivisible objects model. Indeed, the key feature of this model turns out to be that it is always possible not to assign any object to an agent (while, maybe, compensating her for not receiving anything). Therefore, there exists a non-degenerate part of agents' consumption set their preferences over which always matter. The result says the following: efficiency, independence and consistency force us to evaluate allocations by focusing on money equivalents, that is, on the quantities of money that agents deem equivalent to receiving their assigned bundles. It turns out that money equivalents information is sufficient to build complete rankings of feasible allocations, whatever those allocations are.

Our second and main result shows that it is possible to use fairness properties to select among all possible rankings based on money equivalents. But it turns out that not all degree of inequality aversion is finally allowed. Indeed, the only social ordering function which is compatible with

an anonymity axiom and an appropriate version of the Pigou–Dalton principle of transfer¹ turns out to be the money equivalent leximin function (according to the leximin, a vector is preferred to another one if the smallest element of the former is larger than that of the latter, or, in case of a tie, if the second smallest element is larger, and so on).

One core issue in welfare economics has traditionally been the design of social objectives suitable for applications in resource allocation (or reallocation) problems. By mainly uncovering impossibilities, the theory of social choice in economic environment has failed to offer suitable objectives. By focusing on allocation rules, the theory of fair allocation has often offered objectives that can be implemented only if the information on preferences is complete among agents. By combining those approaches, this paper proposes a way out of the dilemma. This new approach has been applied to other contexts, and has also been used to provide solutions to problems of interest to public economics (see, e.g., Fleurbaey [6], Fleurbaey and Maniquet [9] and [10]).

The paper is organized as follows. We define the model in Section 2. We define *Efficiency*, *Independence of Preferences over Infeasible Bundles* and *Consistency* in Section 3, where we also develop and discuss our first result. Then, we define the fairness axioms in Section 4 and prove our main result. We conclude the paper in Section 5.

2. The model

Let us begin with an example. A university department has a set of housings on campus to allocate to visitors. There are more visitors than housings, and no rights have been a priori allocated to visitors. Housings have to be allocated, and visitors who are not assigned campus housing should be given compensation or a subsidy to find housing elsewhere. All visitors would prefer to be housed on campus. The process of allocating housings should depend on visitors' preferences over them. If this process consists of defining a complete ranking on possible ways of allocating housings and organizing transfers among visitors, then the department will be able to deal with feasibility constraints such as visitor i was already present last year and should not be asked to move, visitor j suffers from allergies and should not be sent to housing a , which is too close to the forest, housing b needs repair and is no longer available, etc. Moreover and most importantly, when the assignment process is chosen, if it turns out that visitors are likely to manipulate information regarding their preferences, then the department has to design a revelation mechanism and may choose among the available mechanisms the one that is likely to yield the highest possible assignment in the ranking.

There is an infinite set $\mathcal{A}$ of objects, and an infinite set $\mathcal{N}$ of agents. In specific economies, agents may be assigned either an available object from $\mathcal{A}$ or no object at all. In the latter case, we say that this agent receives the “null object,” which is denoted v . Let $\mathcal{A}^* \equiv \mathcal{A} \cup \{v\}$. An *economy* is a list $E = (N, A, R)$ consisting of a finite set of agents $N \subset \mathcal{N}$, a finite set of objects $A \subset \mathcal{A}$, and a finite list of preferences $R = (R_i)_{i \in N}$ defined on $\mathcal{A}^* \times \mathbb{R}$. We assume that there are at least two agents and that there is at least one object to assign but never more objects than agents, that is, $\#N \geq 2$ and $1 \leq \#A \leq \#N$. Let I_i be the indifference relation associated to R_i , and P_i the strict preference relation. We assume that preferences are continuous and strictly monotone with respect to money. We also assume that all objects are desirable and their values is always

¹ This principle says that a money transfer from a richer agent to a poorer one decreases inequality (provided the transfer does not reverse the wealth ordering of the two agents). In the one-dimensional income inequality measurement framework, this principle is compatible with any degree of inequality aversion (see, e.g., Chakravarty [4]).

finite, that is, for all $(a_i, m_i) \in \mathcal{A} \times \mathbb{R}$, $(a_i, m_i) P_i(v, m_i)$, and there exists $m'_i \in \mathbb{R}$, such that $(a_i, m_i) I_i(v, m'_i)$. Let $\mathcal{R}$ denote the set of all such preference relations. Let $\mathcal{E}$ denote the set of all such economies.

A bundle for an agent $i \in N$ in an economy $E = (N, A, R) \in \mathcal{E}$ is a pair $z_i = (a_i, m_i) \in A \cup \{v\} \times \mathbb{R}$. A feasible allocation for an economy $E = (N, A, R) \in \mathcal{E}$ is a list $z = (z_i)_{i \in N}$ such that no two agents are assigned the same “real” object, all assignments are from the set of available objects, and the sum of monetary compensations does not exceed 0. Let $Z(E)$ denote the set of feasible allocations for E , that is, $z = ((a_i, m_i))_{i \in N} \in Z(E)$ if and only if

$$a_j = a_k \Rightarrow a_j = v = a_k,$$

$$\bigcup_{i \in N} a_i \subseteq A \cup \{v\},$$

$$\sum_{i \in N} m_i \leq 0.$$

Feasible allocations are thus defined in a way which allows us to assign only a subset of the available objects. We come back to this assumption in the conclusion. For an economy $E = (N, A, R) \in \mathcal{E}$, for a feasible allocation $z \in Z(E)$, for $M \subset N$, we write R_M to denote the restriction of R to members of M , and z_M to denote the similar restriction of z .

A social ordering for an economy $E = (N, A, R) \in \mathcal{E}$ is a complete, reflexive and transitive ranking of the feasible allocations. A social ordering function $\mathbf{R}$ associates to each economy $E \in \mathcal{E}$ a social ordering $\mathbf{R}(E)$ for this economy. Let $\mathbf{I}$ be the indifference relation associated to $\mathbf{R}$, and $\mathbf{P}$ the strict preference relation.

3. Independence axiom and the money equivalent property

In order to define social ordering functions, we impose axioms capturing some desirable property for such functions. Our first main axiom is borrowed from the (Arrovian) social choice theory on economic domains. It was introduced by Plott [19] in the abstract framework, and later applied to economic domains (see Le Breton and Weymark [13]). It requires that changes in agents' preferences that leave unaffected their preferences over feasible bundles should not affect the social ordering. Formally,

Independence of Preferences over Infeasible Bundles. For all $E = (N, A, R)$, $E' = (N, A, R') \in \mathcal{E}$, if for all $i \in N$, $a, a' \in A \cup \{v\}$, $m, m' \in \mathbb{R}$,

$$(a, m) R_i(a', m') \Leftrightarrow (a, m) R'_i(a', m'),$$

then for all $z, z' \in Z(E)$,

$$z \mathbf{R}(E) z' \Leftrightarrow z \mathbf{R}(E') z'.$$

This axiom can also be justified from an implementation point of view. The only information that can be revealed by agents' actions are, obviously, information about feasible bundles. The above independence axiom is, therefore, a necessary condition for (any kind of) implementation.

Our second main axiom is inspired by the Consistency property which has been extensively studied in the fair allocation literature (see Thomson [24]). Consistency usually applies to allocation rules. It requires that if an allocation is selected for an economy, then the suballocation

obtained by removing a subset of agents with their assignments of good and money be also selected for the subeconomy (see Tadenuma and Thomson [22]). Adapting this property in our setting, we obtain the following *Consistency* axiom. If an allocation is socially as good as another, and if a subset of agents are assigned exactly the same bundles in both allocations, then removing those agents with their assignments should not change the social preference, that is, the suballocation obtained from the first allocation by removing those agents should still be as good for the subeconomy as the suballocation obtained from the second allocation, provided those suballocations are feasible in the subeconomy. Formally,

Consistency. For all $E = (N, A, R) \in \mathcal{E}$, $M \subset N$, $z, z' \in Z(E)$, if $z_i = z'_i$ for all $i \in N \setminus M$, then

$$z\mathbf{R}(E)z' \Rightarrow z_M\mathbf{R}\left(M, A \setminus \bigcup_{i \in N \setminus M} \{a_i\}, R_M\right)z'_M, \quad \text{and}$$

$$z\mathbf{P}(E)z' \Rightarrow z_M\mathbf{P}\left(M, A \setminus \bigcup_{i \in N \setminus M} \{a_i\}, R_M\right)z'_M,$$

provided $z_M, z'_M \in Z(M, A \setminus \bigcup_{i \in N \setminus M} \{a_i\}, R_M)$.

Consistency requires that the social evaluation between two allocations be unaffected by the removal of agents receiving the same bundles in both allocations. We may think of the converse property that social evaluation be unaffected by the addition of agents receiving the same bundles. It turns out that *Consistency* is equivalent to such a *Consistency Under Addition* property.

Consistency Under Addition. For all $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, $\tilde{E} = (\tilde{N}, \tilde{A}, \tilde{R}) \in \mathcal{E}$ such that $N \cap \tilde{N} = \emptyset$ and $A \cap \tilde{A}, \tilde{z} \in Z(\tilde{E})$,

$$z\mathbf{R}(E)z' \Rightarrow (z, \tilde{z})\mathbf{R}(E, \tilde{E})(z', \tilde{z}), \quad \text{and}$$

$$z\mathbf{P}(E)z' \Rightarrow (z, \tilde{z})\mathbf{P}(E, \tilde{E})(z', \tilde{z}).$$

Let us sketch the easy proof of the equivalence between these two properties. Suppose the first relation of the definition of *Consistency Under Addition* is wrong, that is, $z\mathbf{R}(E)z'$, whereas $(z', \tilde{z})\mathbf{P}(E, \tilde{E})(z, \tilde{z})$. A similar proof works for the second relation (as well as for the statement that *Consistency Under Addition* implies *Consistency*). By *Consistency*, $(z', \tilde{z})\mathbf{P}(E, \tilde{E})(z, \tilde{z})$ implies that $z'\mathbf{P}(E)z$, the desired contradiction.

In the literature on allocation rules, an axiom of *Converse Consistency* has been studied (see Thomson [25]). It states that, in a given economy, if any suballocation of an allocation is in the selection operated by the allocation rule in the corresponding subeconomy, then this allocation must be in the selection for the original economy. Let us observe that our *Consistency Under Addition* property, contrary to *Converse Consistency*, allows us to extend a social evaluation from smaller to larger economies obtained by merely adding new agents. It proves that even if these axioms capture similar requirements, they may have quite different logical implications.

Nothing prevents us up to now from assigning the objects independently of preferences. We therefore impose Paretian type axioms. We stick to the traditional *Strong Pareto* requirement and its weak consequence of *Pareto Indifference*. *Strong Pareto* requires that if each agent weakly prefers her assigned bundle in one allocation over that in another, then the former allocation be also weakly preferred by the society to the latter. If, in addition, at least one agent strictly prefers

the former allocation, then it is also strictly preferred. *Strong Pareto* will only be used in the next section, but we define it here, due to its logical relationship with *Pareto Indifference*, which will play a crucial role in our first result.

Strong Pareto. For all $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, if $z_i R_i z'_i$ for all $i \in N$, then $z \mathbf{R}(E) z'$. If, in addition, $z_j P_j z'_j$ for some $j \in N$, then $z \mathbf{P}(E) z'$.

Pareto Indifference requires that two allocations which are deemed equally good by all agents be also viewed socially equivalent.

Pareto Indifference. For all $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, if $z_i I_i z'_i$ for all $i \in N$, then $z \mathbf{I}(E) z'$.

In our first result, we characterize the class of social welfare functions which satisfy our main axioms, *Independence of Preferences over Infeasible Bundles* and *Consistency*, together with *Pareto Indifference*. If a social ordering function satisfies those three axioms, then it also satisfies the property that in order to evaluate two allocations, only the quantities of money leaving the agents indifferent between their assigned bundles and not receiving any object matters. This quantity of money is called the money equivalent of the corresponding bundle. That only money equivalent matters comes from the fact that, since it is always possible to assign the null object to an agent, the part of the consumption set where money equivalents are computed is always part of the set of feasible bundles, and, more precisely, is the proper intersection of all the possible sets of feasible bundles. By the assumption that the values of the objects are always finite, it turns out that the information given by the money equivalents is all that matters for the construction of social ordering functions.

Money Equivalent Property. For all $E = (N, A, R), E' = (N, A', R') \in \mathcal{E}$, for all $z, z' \in Z(E), y, y' \in Z(E')$, if there exist $m^*, m^{*'} \in \mathbb{R}^N$ such that

$$z_i I_i(v, m_i^*) I'_i y_i \quad \text{and} \quad z'_i I_i(v, m_i^{*'}) I'_i y'_i \quad \text{for all } i \in N,$$

then

$$z \mathbf{R}(E) z' \Leftrightarrow y \mathbf{R}(E') y'.$$

Money equivalents provide numerical representations of the preferences. Formally, the *money utility function* $u: \mathcal{R} \times (\mathcal{A}^* \times \mathbb{R}) \rightarrow \mathbb{R}$ is defined by $u(R_i, z_i) = m \Leftrightarrow z_i I_i(v, m)$.

Lemma 1. *If a social ordering function satisfies the Money Equivalent Property, then it satisfies Pareto Indifference and Independence of Preferences over Infeasible Bundles. Conversely, if a social ordering function $\mathbf{R}$ satisfies Pareto Indifference, Independence of Preferences over Infeasible Bundles, and Consistency, then it satisfies the Money Equivalent Property.*

Proof. The simple proof of the first statement is omitted. Let us suppose that $\mathbf{R}$ satisfies *Pareto Indifference*, *Independence of Preferences over Infeasible Bundles*, and *Consistency*. By *Consistency*, $\mathbf{R}$ also satisfies *Consistency Under Addition*. Let $E = (N, A, R), E' = (N, A', R') \in \mathcal{E}$, $z, z' \in Z(E)$, and $y, y' \in Z(E')$ be such that

$$u(R_i, z_i) = u(R'_i, y_i) \quad \text{and} \quad u(R_i, z'_i) = u(R'_i, y'_i). \quad (1)$$

Let us assume that

$$z\mathbf{R}(E)z'. \quad (2)$$

We have to prove that $y\mathbf{R}(E')y'$. Let $n = \#N$. We begin by constructing two sets of n bundles which are infeasible for E . Let $\varepsilon > 0$. Let $\tilde{m}, \tilde{m}' \in \mathbb{R}^N$ be defined by

$$\begin{aligned} \tilde{m}_i &= \min\{-\varepsilon, u(R_i, z_i) - \varepsilon\}, \\ \tilde{m}'_i &= \tilde{m}_i \quad \text{if } z_i I_i z'_i, \\ \tilde{m}_i - \varepsilon &\quad \text{if } z_i P_i z'_i, \\ \tilde{m}_i + \varepsilon &\quad \text{if } z'_i P_i z_i. \end{aligned}$$

Let $\tilde{A} \subset \mathcal{A}$ be such that $\tilde{A} \cap (A \cup A') = \emptyset$ and $\#\tilde{A} = n$, so that we can find a bijection $\sigma: N \rightarrow \tilde{A}$. Let $\tilde{N} \subset \mathcal{N}$ be such that $\tilde{N} \cap N = \emptyset$ and $\#\tilde{N} = n$, so that we can find a bijection $\rho: N \rightarrow \tilde{N}$. Let $\tilde{z}, \tilde{z}' \in (\tilde{A} \times \mathbb{R})^{\tilde{N}}$ be defined by: for all $i \in N$,

$$\begin{aligned} \tilde{z}_{\rho(i)} &= (\sigma(i), \tilde{m}_i), \\ \tilde{z}'_{\rho(i)} &= (\sigma(i), \tilde{m}'_i). \end{aligned}$$

Let $\bar{R} \in \mathcal{R}^N$ be such that for all $i \in N, a, a' \in A \cup \{v\}, m, m' \in \mathbb{R}$,

$$(a, m)R_i(a', m') \Leftrightarrow (a, m)\bar{R}_i(a', m'),$$

and for all $i \in N$,

$$z_i \bar{I}_i \tilde{z}_{\rho(i)} \quad \text{and} \quad z'_i \bar{I}_i \tilde{z}'_{\rho(i)}. \quad (3)$$

Given the way $\tilde{z}$ and $\tilde{z}'$ were constructed, such preferences exist. By *Independence of Preferences over Infeasible Bundles*, Eq. (2) implies

$$z\mathbf{R}(N, A, \bar{R})z'. \quad (4)$$

Let $\tilde{E} = (\tilde{N}, \tilde{A}, \tilde{R}) \in \mathcal{E}$ be defined by: for all $i \in N$,

$$\tilde{R}_{\rho(i)} = \bar{R}_i. \quad (5)$$

By *Consistency and Consistency Under Addition*, Eq. (4) implies

$$(z, \tilde{z})\mathbf{R}(N \cup \tilde{N}, A \cup \tilde{A}, (\bar{R}, \tilde{R}))(z', \tilde{z}).$$

By *Pareto Indifference*, and Eqs. (3) and (5),

$$(\tilde{z}, z)\mathbf{R}(N \cup \tilde{N}, A \cup \tilde{A}, (\bar{R}, \tilde{R}))(\tilde{z}', z).$$

By *Consistency*,

$$\tilde{z}\mathbf{R}(N, \tilde{A}, \tilde{R})\tilde{z}'. \quad (6)$$

Let $\bar{\bar{R}} \in \mathcal{R}^N$ be the list of preferences which coincide with $\bar{R}$ on the bundles having an element of $\tilde{A} \cup \{v\}$ as first component, and with $\tilde{R}'$ on the bundles having an element of $A' \cup \{v\}$ as first component, that is, for all $i \in N, a, a' \in \tilde{A} \cup \{v\}, b, b' \in A' \cup \{v\}, m, m' \in \mathbb{R}$,

$$\begin{aligned} (a, m)\bar{\bar{R}}_i(a', m') &\Leftrightarrow (a, m)\bar{R}_i(a', m'), \quad \text{and} \\ (b, m)\bar{\bar{R}}_i(b', m') &\Leftrightarrow (b, m)\tilde{R}'_i(b', m'). \end{aligned} \quad (7)$$

By *Independence of Preferences over Infeasible Bundles*, Eq. (6) implies

$$\tilde{z}\mathbf{R}(N, \tilde{A}, \bar{\bar{R}})\tilde{z}'. \quad (8)$$

Let $\tilde{\tilde{R}} \in \mathcal{R}^{\tilde{N}}$ be defined by for all $i \in N$,

$$\tilde{\tilde{R}}_{\rho(i)} = \bar{\bar{R}}_i. \quad (9)$$

By *Consistency* and *Consistency Under Addition*, Eq. (8) implies

$$(\tilde{z}, y)\mathbf{R}(N \cup \tilde{N}, A' \cup \tilde{A}, (\bar{\bar{R}}, \tilde{\tilde{R}}))(\tilde{z}', y).$$

By *Pareto Indifference* and Eqs. (1), (3), (5) and (9),

$$(y, \tilde{z})\mathbf{R}(N \cup \tilde{N}, A' \cup \tilde{A}, (\bar{\bar{R}}, \tilde{\tilde{R}}))(y', \tilde{z}).$$

By *Consistency*,

$$y\mathbf{R}(N, A', \bar{\bar{R}})y'.$$

By *Independence of Preferences over Infeasible Bundles* and Eq. (7),

$$y\mathbf{R}(E')y',$$

the desired outcome. $\square$

Before introducing fairness properties, let us discuss the result presented in Lemma 1. Welfarism is the view that individual utilities are all that matters for equitable social decision-making. Resource allocation decisions should all be reached by considering their impact on individual utilities. Once impacts on utilities are determined, choices are made on ground of an aggregation rule of individual utilities. This aggregation rule does not depend on the specific choice to be made, nor on the specific utility function of the agents (the literature has extensively studied the plausible aggregation rules; see d'Aspremont and Gevers [5] for a survey). On the other hand, it does not say anything about how (cardinally measurable and comparable) individual utilities should be constructed.

It turns out that the *Money Equivalent Property* is equivalent to welfarism. But, moreover and more importantly, it also tells us how to construct utility representations of the preferences, as we are left with no choice but to aggregate money utilities.

Money Utility Welfarism. For all $N \in \mathcal{N}$, there exists an ordering $\mathbf{R}^N$ on $\mathbb{R}^N$ such that for all $A \subset \mathcal{A}$, $R \in \mathcal{R}^N$ such that $E = (N, A, R) \in \mathcal{E}$, for all $z, z' \in Z(E)$,

$$z\mathbf{R}(E)z' \Leftrightarrow u\mathbf{R}^Nu'$$

where $u_i = u(R_i, z_i)$ and $u'_i = u(R_i, z'_i)$ for all $i \in N$.

Lemma 2. A social ordering function $\mathbf{R}$ satisfies the *Money Equivalent Property* if and only if it satisfies *Money Utility Welfarism*.

Proof. “if”: Let $E = (N, A, R)$, $E' = (N, A', R') \in \mathcal{E}$, $z, z' \in Z(E)$, and $y, y' \in Z(E')$. By *Money Utility Welfarism*, there exists an ordering $\mathbf{R}^N$ on $\mathbb{R}^N$ such that

$$z\mathbf{R}(E)z' \Leftrightarrow u\mathbf{R}^Nu'$$

where $u_i = u(R_i, z_i)$ and $u'_i = u(R'_i, z'_i)$ for all $i \in N$, and

$$y\mathbf{R}(E')y' \Leftrightarrow v\mathbf{R}^N v'$$

where $v_i = u(R'_i, y_i)$ and $v'_i = u(R'_i, y'_i)$ for all $i \in N$. If there exist $m^*, m^{*'} \in \mathbb{R}^N$ such that

$$z_i I_i(v, m_i^*) I'_i y_i \quad \text{and} \quad z'_i I_i(v, m_i^{*'}) I'_i y'_i$$

then for all $i \in N$, $u_i = v_i$ and $u'_i = v'_i$. Gathering the above relations yields

$$z\mathbf{R}(E)z' \Leftrightarrow y\mathbf{R}(E')y'.$$

“only if”: Let $\mathbf{R}$ satisfy the *Money Equivalent Property*. Let us fix $N \in \mathcal{N}$ throughout the proof, and let us call $\mathcal{E}^N$ the set of economies where the set of agents is N . Let $E \in \mathcal{E}^N$. Let $X(E) \subset \mathbb{R}^N$ be defined by:

$$x \in X(E) \Leftrightarrow \exists z \in Z(E) \text{ s.t. } x_i = u(R_i, z_i), \forall i \in N.$$

For $x, x' \in X(E)$, we write $x\mathbf{R}^N x'$ if and only if there exist $z, z' \in Z(E)$ such that for all $i \in N$, $x_i = u(R_i, z_i)$, $x'_i = u(R_i, z'_i)$ and $z\mathbf{R}(E)z'$. By the *Money Equivalent Property*, $\mathbf{R}^N$ is an ordering on $X(E)$. Let $E' \in \mathcal{E}^N$ and let $X(E')$ be defined as above. Let $x, x' \in X(E) \cap X(E')$. Let $z, z' \in Z(E)$, and $y, y' \in Z(E')$ be such that for all $i \in N$, $u(R_i, z_i) = u(R'_i, y_i)$ and $u(R_i, z'_i) = u(R'_i, y'_i)$. By the *Money Equivalent Property*, $z\mathbf{R}(E)z' \Leftrightarrow y\mathbf{R}(E')y'$. This proves that $\mathbf{R}$ is welfarist in its ranking over all allocations having a money utility representation in $X(E)$. Now, E was chosen arbitrarily. So $\mathbf{R}$ is welfarist on $X(E)$ for all $E \in \mathcal{E}^N$. It remains to show that $\bigcup_{E \in \mathcal{E}^N} X(E) = \mathbb{R}^N$. Let $x \in \mathbb{R}^N$. We must construct $E \in \mathcal{E}^N$ such that $x \in X(E)$. For all $i \in N$, let $a_i \in \mathcal{A}$ be such that $a_i \neq a_j$ for all $j \neq i$, and let $R_i \in \mathcal{R}$ be such that

$$x_i \geq 0 \Rightarrow u(R_i, (a_i, 0)) = x_i, \quad \text{and}$$

$$x_i < 0 \Rightarrow u(R_i, (a_i, x_i)) = x_i.$$

Let $A = \bigcup_{i \in N} a_i$. Let $E = (N, A, R)$. By construction, $E \in \mathcal{E}^N$. Let z be defined by: for all $i \in N$, $z_i = (a_i, \min\{0, x_i\})$. By construction, $z \in Z(E)$. Also, $u(R_i, z_i) = x_i$ for all $i \in N$. Therefore, $x \in X(E)$. $\square$

In conclusion, in this specific model of indivisible good assignment with money, combining *Pareto Indifference*, *Independence of Preferences over Infeasible Bundles* and *Consistency* forces us to be welfarist and to use money utility as the proper indicator of individual welfare. But there is no restriction yet on how to aggregate money equivalents. We come to this question in the next section.

4. Fairness and the money equivalent leximin

The fair allocation literature which focuses on allocation rules has proposed a long list of equity axioms (see Moulin and Thomson [17]). One of the basic axioms, often called *Equal Treatment of Equals*, requires that if two agents have the same preferences, then they be assigned the same bundle, or at least, bundles they deem equivalent. Here we consider two possible adaptations of this requirement to our current framework. The first one is borrowed from Fleurbaey [6]. It refers to the Pigou–Dalton principle of transfer. This principle is at the heart of inequality measurement theory. It requires that an income transfer from an agent to a poorer one reduce inequality, as long as the income ranking of those two agents remains unaffected. That principle

clearly favors equality but is consistent with any degree of inequality aversion. The Fleurbaey generalization of this principle to multi-dimensional framework like ours requires that if two agents have the same preferences and are assigned bundles they do not deem equivalent, then a money transfer from the agent having been assigned the bundle they both prefer to the other agent be viewed a strict social improvement, provided there is no reversal in the indifference curve ranking, that is, provided both agents still consider the final bundle assigned to the first agent at least as good as that assigned to the second one.²

Transfer Principle Among Equals. For all $E = (N, A, R) \in \mathcal{E}$, $z = ((a_i, m_i))_{i \in N}$, $z' = ((a'_i, m'_i))_{i \in N} \in Z(E)$, if there exist $j, k \in N$ such that $R_j = R_k$ and for all $i \neq j, k$, $z_i = z'_i$, then for all $\Delta > 0$,

$$[(a'_j, m'_j) = (a_j, m_j - \Delta) R_{j,k} (a'_k, m'_k) = (a_k, m_k + \Delta)] \Rightarrow [z' \mathbf{P}(E) z].$$

Our second fairness axiom captures the anonymity content of the classical *Equal Treatment of Equals* axiom. It requires that if two agents have the same preferences, then permuting the two bundles they are assigned in an allocation yields another allocation society deems equivalent. This requirement is also compatible with any degree of inequality aversion.³

Anonymity Among Equals. For all $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, if there exist $j, k \in N$ such that $R_j = R_k$ and for all $i \neq j, k$, $z_i = z'_i$, then

$$[z_j = z'_k \text{ and } z'_j = z_k] \Rightarrow [z \mathbf{I}(E) z'].$$

The main result of the paper, which we state and prove at the end of this section, shows that the combination of these fairness axioms with our other axioms leads us to be infinitely averse to money equivalent inequality. Infinite inequality aversion is captured by the so-called leximin, or lexicographic maximin, aggregation criterion. For $m, m' \in \mathbb{R}^N$, we write $m \geq_{\text{lex}} m'$ if and only if the smallest element of m is greater than the smallest element of m' , or they are equal but the second smallest element of m is greater than the second smallest element of m' , and so on.

Definition 1. The *Money Equivalent Leximin* function $\mathbf{R}^L$ works as follows: For all $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, let $m^*, m^{*'} \in \mathbb{R}^N$ be such that

$$z_i I_i(v, m_i^*) \quad \text{and} \quad z'_i I_i(v, m_i^{*'});$$

then

$$z \mathbf{R}^L(E) z' \Leftrightarrow m^* \geq_{\text{lex}} m^{*'}.$$

As we want to insist in this paper on positive results, we begin by proving that any combination of axioms defined so far leads to some possibility. The *Money Equivalent Leximin* function, indeed, satisfies all our axioms.

² If the proviso that agents' preferences are identical is removed, and if no other restriction is imposed on the preferences of those two agents, then the resulting (much stronger) axiom turns out to be incompatible with *Pareto Indifference*. See Fleurbaey and Trannoy [11] for a proof and discussion.

³ It is even compatible with a negative degree of inequality aversion.

Lemma 3. *The Money Equivalent Leximin function $\mathbf{R}^L$ satisfies Strong Pareto, Independence of Preferences over Infeasible Bundles, Consistency, the Transfer Principle among Equals and Anonymity among Equals.*

Proof. To simplify the notation, we use the money utility function u introduced in Section 3. (1) *Strong Pareto:* This comes from the fact that for all $m, m' \in \mathbb{R}^N$, if $m \geq m'$, then $m \geq_{\text{lex}} m'$, and, if, in addition, $m_j > m'_j$ for some $j \in N$, then $m >_{\text{lex}} m'$. (2) *Independence of Preferences over Infeasible Bundles:* Let $E = (N, A, R)$, $E' = (N, A, R') \in \mathcal{E}$, be such that for all $i \in N$, $a, a' \in A \cup \{v\}$, $m, m' \in \mathbb{R}$, $(a, m)R_i(a', m') \Leftrightarrow (a, m)R'_i(a', m')$. Then for all $z, z' \in Z(E)$, and all $i \in N$, we have $u(R_i, z_i) = u(R'_i, z_i)$ and $u(R_i, z'_i) = u(R'_i, z'_i)$, so that $z\mathbf{R}^L(E)z' \Leftrightarrow z\mathbf{R}^L(E')z'$. (3) *Consistency:* This comes from the fact that for all $m, m' \in \mathbb{R}^N$, $M \subset N$, if $m_i = m'_i$ for all $i \in N \setminus M$, then $m \geq_{\text{lex}} m' \Rightarrow m_M \geq m'_M$ and $m >_{\text{lex}} m' \Rightarrow m_M > m'_M$.⁴ (4) *Transfer Principle among Equals:* Let $E = (N, A, R) \in \mathcal{E}$, $z = ((a_i, m_i))_{i \in N}$, $z' = ((a'_i, m'_i))_{i \in N} \in Z(E)$, be such that for some $j, k \in N$, $R_j = R_k$ and for all $i \neq j, k$, $z_i = z'_i$, and, moreover,

$$[(a'_j, m'_j) = (a_j, m_j - \Delta)R_{j,k}(a'_k, m'_k) = (a_k, m_k + \Delta)].$$

Then, $u(R_j, (a_j, m_j)) > u(R_j, (a'_j, m'_j)) \geq u(R_k, (a'_k, m'_k)) > u(R_k, (a_k, m_k))$, so that $z'\mathbf{P}^L(E)z$. (5) *Anonymity among Equals:* Let $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, and $j, k \in N$ be such that $R_j = R_k$, for all $i \neq j, k$, $z_i = z'_i$, and $z_j = z'_k$ and $z'_j = z_k$. Then $u(R_j, z_j) = u(R_k, z'_k)$ and $u(R_j, z'_j) = u(R_k, z_k)$, so that $z\mathbf{I}^L(E)z'$. $\square$

Now, we pursue our analysis of money utility by showing that the fairness axioms defined above, when combined with *Strong Pareto* and the *Money Equivalent Property* force us to be infinitely averse to money utility inequality, that is, characterize the *Money Equivalent Leximin* function $\mathbf{R}^L$. *Transfer Principle among Equals* yields the maximin property in money equivalent, whereas *Anonymity among Equals* implies that the maximin be applied lexicographically. This is made clear through the following two lemmas.

Money Equivalent Maximin Property. For all $E = (N, A, R) \in \mathcal{E}$, for all $z, z' \in Z(E)$, if $m^*, m^{*'} \in \mathbb{R}^N$ are such that $z_I I_i(v, m_i^*)$ and $z'_I I_i(v, m_i^{*'})$ then

$$\min_{i \in N} \{m_i^*\} > \min_{i \in N} \{m_i^{*'}\} \Rightarrow z\mathbf{P}(E)z'.$$

Let us notice that the above property does not imply the Money Equivalent Property.

Lemma 4. *If a social ordering function $\mathbf{R}$ satisfies the Strong Pareto, Money Equivalent Property, and the Transfer Principle among Equals, then it satisfies the Money Equivalent Maximin Property in all $E = (N, A, R) \in \mathcal{E}$ such that $\#A \geq 2$.*

Proof. Again, we use the u function terminology. Let $E = (N, A, R) \in \mathcal{E}$, $z, z' \in Z(E)$, be such that

$$\min_{i \in N} \{u(R_i, z_i)\} > \min_{i \in N} \{u(R_i, z'_i)\}.$$

⁴ Our conventions for vector inequalities: for $x, y \in \mathbb{R}^L$, $x \geq y \Leftrightarrow x_l \geq y_l, \forall l \in L$, and $x > y \Leftrightarrow x \geq y$ and $x \neq y$.

Assume, contrary to what needs to be proven, that $z'\mathbf{R}(E)z$. Let $E' = (N, A, R') \in \mathcal{E}$ be such that for all $j, k \in N$, $R'_j = R'_k$ and R'_j has the property that for all $a \in A$, $m, m' \in \mathbb{R}$, and for all $i \in N$,

$$(a, m)I_i(v, m') \Rightarrow (a, m)R'_j(v, m'),$$

which means that the willingness to pay for any object in A is greater for R'_j than for any R_i . Given the restriction on R'_j , there exist $y, y' \in Z(E')$ such that

$$u(R_i, z_i) = u(R'_i, y_i) \quad \text{and} \quad u(R_i, z'_i) = u(R'_i, y'_i). \quad (10)$$

By the *Money Equivalent Property*, Eq. (10) implies

$$y'\mathbf{R}(E')y. \quad (11)$$

Let $j \in N$ be such that $u(R'_j, y'_j) = \min_{i \in N} u(R'_i, y'_i)$. Let N be partitioned into N_1 , and N_2 such that $\#N_1 = n_1$, $\#N_2 = n_2$ and

$$\forall i \in N_1: \quad u(R'_i, y_i) \geq u(R'_i, y'_i),$$

$$\forall i \in N_2: \quad u(R'_i, y_i) < u(R'_i, y'_i).$$

Note that $j \in N_1$. If $N_2 = \emptyset$, then, by *Strong Pareto*, $y'\mathbf{R}(E')y'$, a contradiction. So, let us assume that $N_2 \neq \emptyset$. The remainder of the proof consists in showing that it is possible to build a new allocation y'' , such that $y''\mathbf{P}(E')y$ and N can still be partitioned into two sets, N'_1 and N'_2 , such that $\#N'_1 = n_1 + 1$ and $\#N'_2 = n_2 - 1$. Repeating the argument n_2 times eventually yield the contradiction with *Strong Pareto*. Note that each repetition of the argument typically requires that new preferences be defined, which, by the *Money Equivalent Property*, is always possible. Let $k \in N_2$. Let u''_j, u''_k be such that

$$u(R'_j, y'_j) < u''_j \leq u''_k < u(R'_l, y_l), \quad \forall l = j, k.$$

Let $a, b \in A$. We may and do assume that $a \neq b$, since $\#A \geq 2$. Moreover, R'_j could have been defined in such a way that there is some $\Delta > 0$, and $m_j, m_k \in \mathbf{R}$, such that

$$y'_j I'_j(a, m_j),$$

$$(v, u''_j) I'_j(a, m_j + \Delta),$$

$$(v, u''_k) I'_k(b, m_k - \Delta),$$

$$y'_k I'_k(b, m_k).$$

We do assume that R'_j has this property. Let $\bar{y}, y'' \in Z(E')$ be such that for all $i \neq j, k$, $\bar{y}_i = y''_i I'_i y'_i$, $\bar{y}_j = (a, m_j)$, $\bar{y}_k = (b, m_k)$, $y''_j = (a, m_j + \Delta)$, and $y''_k = (b, m_k - \Delta)$. By *Pareto Indifference*,

$$y'\mathbf{I}(E')\bar{y}. \quad (12)$$

By the *Transfer Principle among Equals*,

$$y''\mathbf{P}(E')\bar{y}. \quad (13)$$

By Eqs. (11), (12) and (13),

$$y''\mathbf{P}(E')y,$$

the desired outcome. $\square$

There is an interesting relationship between this lemma and results in Fleurbaey [6] and Maniquet and Sprumont [14,15]. In those three papers, indeed, fairness axioms a priori consistent with any degree of inequality aversion also yield maximin properties when they are combined with, typically, independence axioms. This common feature of the domains where social ordering functions have been studied up to now stands in contrast to the income inequality measurement framework as well as the social welfare functionals framework. In the former framework, the Pigou–Dalton principle of transfer, even when combined with different types of independence and consistency properties do not yield the maximin property (see Chakravarty [4]). In the latter framework, the traditional equity axiom that leads to the maximin axiom, that is, the Hammond equity axiom, excludes by itself any finite degree of utility inequality aversion (cf. Hammond [12]; see also d’Aspremont and Gevers [5] for a survey).

Lemma 5. *If a social ordering function satisfies the Money Equivalent Property, the Maximin Money Equivalent Property, Consistency and Anonymity among Equals, then it coincides with the Money Equivalent Leximin function $\mathbf{R}^L$ in all $E = (N, A, R) \in \mathcal{E}$ such that $\#A \geq 2$.*

Proof. First, let us recall that the *Money Equivalent Property* implies *Pareto Indifference*. Let $\mathbf{R}$ satisfy the axioms. Let $E = (N, A, R) \in \mathcal{E}$ be such that $\#A \geq 2$. By the *Money Equivalent Property*, we can assume, w.l.o.g., that for all $i, j \in N$, $R_i = R_j$. (If this condition were not satisfied, then we could change the actual profile of preferences into a new profile satisfying the condition, without affecting the social preferences over money equivalent vectors, as we did at the beginning of the proof of Lemma 4.) Let $z = ((a_i, m_i))_{i \in N}$, $z' = ((a'_i, m'_i))_{i \in N} \in Z(E)$. We distinguish two cases.

Case 1. $z \mathbf{I}^L(E) z'$. Let $n = \#N$. Let $\sigma : N \rightarrow \{1, \dots, n\}$ be a bijection satisfying the property that for all $i, j \in N$, $\sigma(i) \geq \sigma(j) \Rightarrow u(R_i, z_i) \geq u(R_j, z_j)$. Let $z_\sigma \in Z(E)$ denote the allocation obtained from z by permuting its component according to σ . Let σ' denote the similar bijection associated to z' , and $z'_{\sigma'}$ the resulting allocation. By *Anonymity among Equals*, $z \mathbf{I}(E) z_\sigma$ and $z' \mathbf{I}(E) z'_{\sigma'}$. By construction, and given that $z \mathbf{I}^L(E) z'$, $z_{\sigma i} I_i z'_{\sigma' i}$ so that by *Pareto Indifference*, $z_\sigma \mathbf{I}(E) z'_{\sigma'}$. Gathering all the above social indifferences yield $z \mathbf{I}(E) z'$, the desired outcome.

Case 2. $z \mathbf{P}^L(E) z'$. Assume, contrary to the statement we have to prove, that

$$z' \mathbf{R}(E) z. \quad (14)$$

Let $n = \#N$. Let $\sigma, \sigma', z_\sigma, z'_{\sigma'}$ be defined as above. By *Anonymity among Equals*, $z \mathbf{I}(E) z_\sigma$ and $z' \mathbf{I}(E) z'_{\sigma'}$. Therefore, by Eq. (14),

$$z'_{\sigma'} \mathbf{R}(E) z_\sigma. \quad (15)$$

Given that $z \mathbf{P}^L(E) z'$, there is $j \in N$ such that for all $i \in N$ such that $\sigma(i) < \sigma(j)$, $z_{\sigma i} I_i z'_{\sigma' i}$ and $z_{\sigma j} P_j z'_{\sigma' j}$. Then, N can be partitioned into N_1, N_2 and N_3 such that $\#N_1 = n_1$, $\#N_2 = n_2$, $\#N_3 = n_3$ and $j = n_1 + 1$, and

$$\begin{aligned} \forall i \in N_1: & \quad \sigma(i) \leq n_1 \quad \text{and} \quad u(R_i, z_{\sigma i}) = u(R_i, z'_{\sigma' i}), \\ \forall i \in N_2: & \quad u(R_i, z_{\sigma i}) > u(R_i, z'_{\sigma' i}), \\ \forall i \in N_3: & \quad u(R_i, z_{\sigma i}) \leq u(R_i, z'_{\sigma' i}). \end{aligned}$$

If $N_1 = \emptyset$, then by the *Maximin Money Equivalent Property*, $z_\sigma \mathbf{P}(E) z'_{\sigma'}$, a contradiction. So let us assume that $N_1 \neq \emptyset$. Our strategy consists in using *Consistency* to remove agents in N_1 from

the economy, so that agent j has the smallest money equivalent, in contradiction to the *Maximin Money Equivalent Property*, but removing those agents may yield an infeasible allocation (not enough money would be left). Let $M = \min\{\sum_{i \in N_1} m_{\sigma i}, \sum_{i \in N_1} m'_{\sigma' i}\}$. Let $\tilde{N} \subset N$, $\tilde{A} \subset A$ be such that $\tilde{N} \cap N = \emptyset$, $\#\tilde{N} = n$, $\tilde{A} \cap A = \emptyset$ and $\#\tilde{A} = n$. Let $\tilde{z} = ((\tilde{a}_i, \tilde{m}_i))_{i \in \tilde{N}} \in (\mathcal{A}^* \times \mathbb{R})^{\tilde{N}}$, be such that $\sum_{i \in \tilde{N}} \tilde{m}_i < M$. Let $\tilde{E} = (\tilde{N}, \tilde{A}, \tilde{R}) \in \mathcal{E}$ be such that for all $i \in \tilde{N}$, $\tilde{R}_i = R_j$. By the *Money Equivalent Property*, we can assume, w.l.o.g., that for all $i \in \tilde{N}$,

$$u(\tilde{R}_i, \tilde{z}_i) > u(R_j, z'_{\sigma' j}).$$

(If this condition were not satisfied, then we could change the actual profile of preferences into a new profile satisfying the condition, without affecting the social preferences over money equivalent vectors.) Let $\bar{z}, \bar{z}' \in Z(E)$ be defined by

$$\begin{aligned} \forall i \in N_2 \cup N_3, \quad \bar{z}_i &= z_{\sigma i} \quad \text{and} \quad \bar{z}'_i = z'_{\sigma' i}, \\ \forall i \in N_1, \quad \bar{z}_i &= \bar{z}'_i = z_{\sigma i} \quad \text{if} \quad \sum_{i \in N_1} m_{\sigma i} \leq \sum_{i \in N_1} m'_{\sigma' i}, \\ & z'_{\sigma' i} \quad \text{if} \quad \sum_{i \in N_1} m_{\sigma i} > \sum_{i \in N_1} m'_{\sigma' i}. \end{aligned}$$

By *Pareto Indifference*, $z_{\sigma} \mathbf{I}(E) \bar{z}$ and $z'_{\sigma'} \mathbf{I}(E) \bar{z}'$, so that Eq. (15) implies

$$\bar{z}' \mathbf{R}(E) \bar{z}.$$

By *Consistency* and *Consistency Under Addition* (remember that *Consistency* is equivalent to *Consistency Under Addition*),

$$(\bar{z}', \bar{z}) \mathbf{R}(E, \tilde{E})(\bar{z}, \bar{z}).$$

By *Consistency*,

$$(\bar{z}'_{N \setminus N_1}, \bar{z}) \mathbf{R} \left((N \setminus N_1) \cup \tilde{N}, \left(A \setminus \bigcup_{i \in N_1} \{\bar{a}_i\} \right) \cup \tilde{A}, (R_{N \setminus N_1}, \tilde{R}) \right) (\bar{z}_{N \setminus N_1}, \bar{z}),$$

which contradicts the *Maximin Money Equivalent Property*, as agent j now has the smallest money equivalent in the allocation $(\bar{z}'_{N \setminus N_1}, \bar{z})$. $\square$

We can gather the results presented so far and state the following theorem.

Theorem 1. *A social ordering function $\mathbf{R}$ satisfies Strong Pareto, Independence of Preferences over Infeasible Bundles, Consistency, the Transfer Principle for Equals, and Anonymity among Equals if and only if it is the Money Equivalent Leximin function $\mathbf{R}^L$.*

Proof. The “if” part was proven in Lemma 3. The “only if” was proven in Lemmas 1, 4 and 5 in the case of economies $E = (N, A, R) \in \mathcal{E}$ such that $\#A \geq 2$. We only need to get rid of this last restriction. But by *Consistency* and *Consistency Under Addition*, we can always go from one object economies to two or more objects economies. Lemma 4 is therefore valid on the entire domain $\mathcal{E}$, and so is Lemma 5. $\square$

Let us note that in a first best context, the Money Equivalent Maximin property imply that only allocations where the Money Equivalents are equalized among agents be selected. Consequently, differences between the Maximin and the Leximin can only arise in second best contexts.

Let us check now that none of the above axioms is redundant.

Let $a^* \in \mathcal{A}$. Let numerical representation of preferences v be defined by

$$v(R_i, z_i) = m \Leftrightarrow (a^*, m)I_i z_i.$$

The v -utility leximin function satisfies all the axioms but *Independence of Preferences over Infeasible Bundles*.

For $a \in \mathcal{A}^*$, let $m^a(R_i, z_i) = m \Leftrightarrow (a, m)I_i z_i$. For all $E = (N, A, R) \in \mathcal{E}$, let the numerical representation of preferences w be defined by

$$w(R_i, z_i, A) = m \Leftrightarrow m = \sum_{a \in A \cup \{v\}} m^a(R_i, z_i).$$

The w -utility leximin function satisfies all the axioms but *Consistency*.

The social ordering function which ranks allocations so as to minimize the variance in money utilities satisfies all the axioms but *Strong Pareto*.

The social ordering function which ranks allocations so as to maximize the sum of agents' money equivalent satisfies all the axioms but the *Transfer Principle for Equals*.

Let $\geq$ denote a complete ordering on $\mathcal{N}$. The social ordering function which coincides with the money equivalent leximin function in case of strict preference, and which prefers, in case of a tie by the leximin, the allocation where the name of agent with smallest money equivalent is the smaller, satisfies all the axioms but *Anonymity among Equals*.

5. Conclusion

The particular feature of the indivisible good assignment model that is exploited in this paper is that it is always feasible not to assign any indivisible good to one agent (that is, assign her only money). This comes from the assumption that we can freely dispose of the objects. Consequently, a share of agents' consumption set always contains feasible bundles and, moreover, is the intersection of all possible feasible sets. This is why the information about which quantity of money would leave an agent indifferent between consuming her actual bundle and not consuming any object but receiving this amount of money is crucial. As it turns out, this information is sufficient to construct social ordering functions.

If we had assumed, however, that there was no free disposal, so that a social ordering only has to rank the allocations where all available objects are allocated, then all the results we obtained would still be valid, provided the number of objects were assumed to be strictly lower than the number of agents.⁵ Under this assumption, indeed, it is always the case that at least one agent does not consume any object, but this agent may be anyone, so that the set of feasible bundles always include the null object bundles.

Our main result depends, on the other hand, on the assumption that we also have to rank allocations where the assigned quantities of money do not sum up to zero. We crucially use that richness of the set of feasible allocations in deriving the maximin property from the transfer axiom. Another richness assumption that is crucial here is the domain of preference richness. If, instead, we had restricted ourselves to quasi-linear preferences, for instance, the maximin property could not have been deduced. Indeed, any generalized utilitarian social welfare function $\sum f(u(R_i, z_i))$ with strictly increasing and concave f would satisfy all our axioms. There are

⁵ This would also require rewriting the *Consistency* property, and adapting the way it is used in the proofs.

other subdomains of economies of interest, however, that would not prevent our results from holding. If, for instance, we restrict our attention to economies where there is an objective ranking of the objects and all agents prefer objects higher in that ranking, then the Utility Equivalent Leximin function remains the only function satisfying our axioms.

Let us conclude by drawing two general lessons from this paper. First of all, our result (together with results in the other recent papers studying fair social orderings) shows that fairness can be studied through the definition of social orderings and not only allocation rules. As explained above, social orderings offer much finer judgments than allocation rules.⁶ On the other hand, allocation rules can easily be deduced from social ordering functions, by simply taking the selection to be the set of socially preferred allocations. The allocation rule associated with the money equivalent leximin function then selects the subset of efficient allocations where all agents have the same money equivalent, that is, they all are indifferent between their assigned bundle and a common null object bundle. Let us call it the *Equal Money Equivalent* rule. It is an egalitarian-equivalent allocation rule, a concept introduced by Pazner and Schmeidler [18].

There is no clear logical relationship, however, between the analysis of social ordering functions and the analysis of allocation rules. The model studied in this paper offers the best example of this fact. For the *Equal Money Equivalent* rule does not rely on preferences over infeasible bundles, is consistent, selects all Pareto indifferent allocations, is Pareto efficient and treats equals equally. In view of Theorem 1, one may therefore wonder if it is the only rule satisfying those requirements. It is not, as proven by the following example. The allocation rule which has, by far, received the largest attention in the literature, is the one selecting the efficient and envy-free allocations (see Alkan [1], Alkan, Demange and Gale [2], Maskin [16], Svensson [21], and Tadenuma and Thomson [22]; an exception is Bevia [3]). An allocation is envy-free if no agent strictly prefers the bundle assigned to any other agent to her own bundle. The efficient and envy-free rule is independent of preferences over infeasible bundles, consistent, and efficient. It treats equals equally, and selects Pareto indifferent allocations. But that rule selects sets of bundles that, in some economies, are disjoint from the set of allocations selected by the *Equal Money Equivalent* rule (this is a corollary of the main result of Thomson [23]).

Our second general lesson is the following. This paper shows that, contrary to the general wisdom that is currently dominant in welfare economics, it is possible to build social orderings in economic domains without relying on interpersonal comparison of utility, so that, in particular, the objective of the social planner in public economics need not be utilitarian. At least, this has been shown in a few models. An urgent task in welfare economics, therefore, is to extend this line of research to other models as well.

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⁶ It is often argued that the success of the theory of fair allocation (in the sense of it uncovering mainly possibility rather than impossibility results) comes from the fact that the attention is put on allocation rules rather than orderings (see e.g. Varian [26], Sen [20] and Moulin and Thomson [17]). It is clear after this paper and the other papers on social ordering functions that the argument does not hold true.

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